

Machine Learning

Evaluation of Classifiers and Basics of Regression

FAST

DISCOVERING
THE FUTURE

Topics of previous lecture

- ✓ Ingredients of Machine Learning
- ✓ Classification Basics
- ✓ Basic Linear Classifier
- ✓ K-Nearest Neighbours Classifier
- ✓ Naive Bayes Classifier
- ✓ Decision Trees
- ✓ Support Vector Machines (SVM)
- ✓ Ensemble Methods (Bagging, Weighted Voting, Stacking)

Topics of today's lecture

- Evaluation and Scoring of Classifiers
- Main Concepts in Regression
- Linear Regression
- Ordinary Least Squares (OLS)

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- Test set error rate can be seen as an estimate of the probability of the model error on a random instance $\mathbf{x} \in \mathcal{X}$ from the instance space

$$\mathbb{P}(\hat{f}(\mathbf{x}) \neq f(\mathbf{x}))$$

Confusion Matrix

- Often it is useful to see the kind of errors that the classifier makes

	Predicted $\oplus$	Predicted $\ominus$	
Actual $\oplus$	30	20	50
Actual $\ominus$	10	40	50
	40	60	100

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- This example of a confusion matrix (also known as contingency table) shows prediction performance on 100 instances

Terminology in binary classification

- The elements in the confusion matrix are called as follows:

	Predicted $\oplus$	Predicted $\ominus$	
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- **F-measure** is the harmonic mean of precision and recall $F = \frac{2 \cdot prec \cdot rec}{prec + rec}$

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- For example in medical diagnostic testing
 - Few disease cases (positive class)
 - Many healthy cases (negative class)

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- A better solution: ask classifiers to output scores:
 - Higher score means more likely positive
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- By choosing the decision threshold we can change trade-off between TPR and FPR

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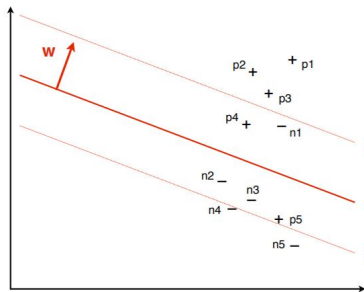
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- RF (random forest):
 - Score = proportion of trees predicting positive

Scoring for SVM



Threshold A: $p1 - p2 - p3 - n1 - p4 - n2 - n3 - p5 - n4 - n5$

Threshold B: $p1 - p2 - p3 - n1 - p4 - n2 - n3 - p5 - n4 - n5$

Threshold C: $p1 - p2 - p3 - n1 - p4 - n2 - n3 - p5 - n4 - n5$

Example

True labels

$(1, 0, 1, 0, 1)$



Classifier predicts

$(0.6, 0.2, 0.7, 0.5, 0.4)$

Example

True labels

$(1, 1, 0, 1, 0)$



$(0.7, 0.6, 0.5, 0.4, 0.2)$

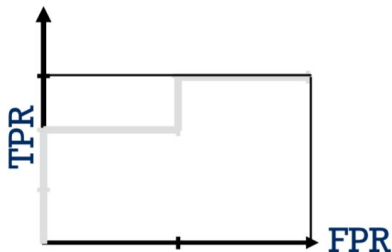
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$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

Navigation icons: back, forward, search, etc.

Example

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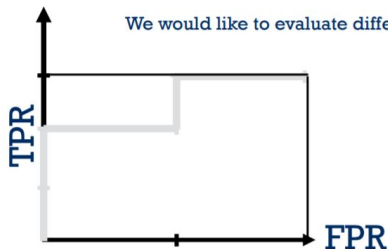
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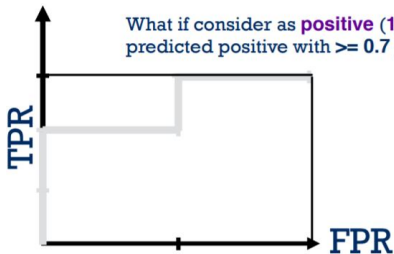
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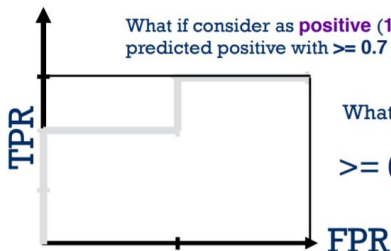
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What would **TPR** and **FPR** be in this case?

$$\geq 0.7 \quad \text{TPR} = 1/3$$
$$\text{FPR} = 0 / (0 + 2)$$

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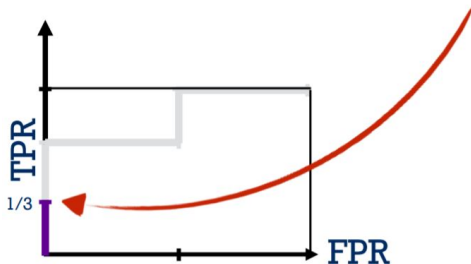
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Let's plot this point on a graph



Example

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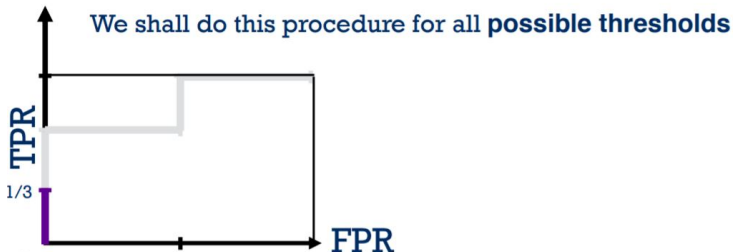


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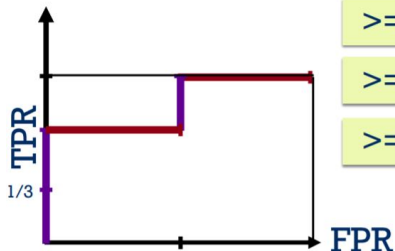
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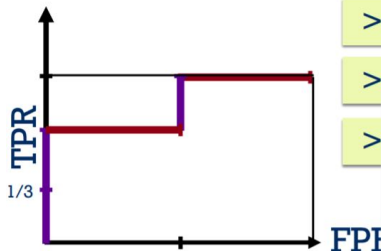
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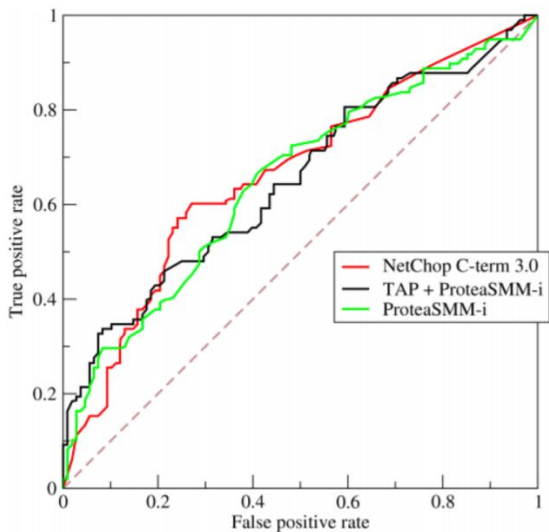
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This curve is called the Receiver Operating Characteristic (ROC) and the area under it is denoted as AUC and for an ideal classifier it should be close to 1

Typical comparison of classifiers



- Regression task is very similar to the classification task, except that we must predict a continuous variable (instead of a categorical class label)

Intro to Regression

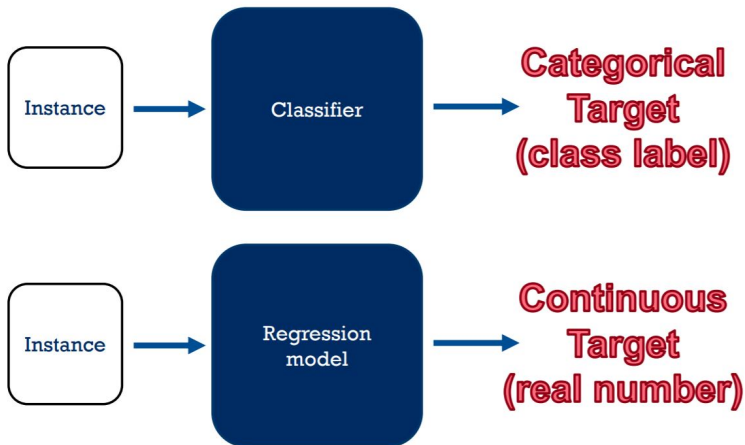
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- For example:
 - predict the salary given the info about a person
 - predict the risk of a policyholder for insurance
 - predict the expected number of days that a patient will stay in a hospital
 - ...

Classification vs Regression model



Mathematical notation of classification

- $\mathbb{X}$ - input space (set of all possible instances)
- $\mathbb{Y}$ - output space (all possible labels)
- $f : \mathbb{X} \rightarrow \mathbb{Y}$ - any such function is a classifier
- $\mathbf{x} \in \mathbb{X}$ - instance
- $y \in \mathbb{Y}$ - actual / true label of instance $\mathbf{x}$
- $\hat{y} = f(\mathbf{x})$ - predicted label of instance $\mathbf{x}$

Mathematical notation of classification regression

- $\mathbb{X}$ - input space (set of all possible instances)
- $\mathbb{Y}$ - output space (all possible labels **real numbers**)
- $f : \mathbb{X} \rightarrow \mathbb{Y}$ - any such function is a classifier **regression model**
- $\mathbf{x} \in \mathbb{X}$ - instance
- $y \in \mathbb{Y}$ - actual / true label **target value** of instance $\mathbf{x}$
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- For this we are given training data $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n) \in \mathbb{X} \times \mathbb{R}$

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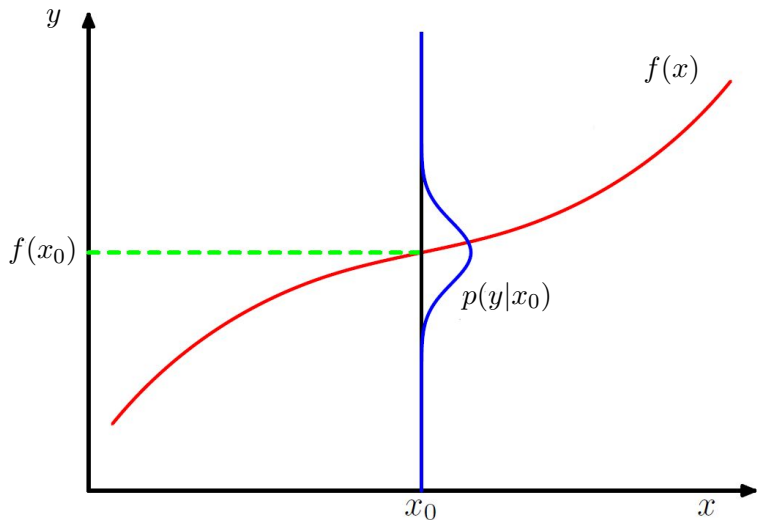
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- What does this really mean?

Learning Problem in Regression



Derivation of the Regression function

Definition

For X and Y continuous random variables, the conditional expectation is

$$\mathbb{E}(X|Y) = \int_{x \in \mathcal{X}} xp(x|y)dx,$$

where $p(x|y) = \frac{p(x,y)}{p(y)}$ is the conditional density function of X given Y .

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- Solving for $f(\mathbf{x})$ gives

$$f(\mathbf{x}) = \frac{\int y p(\mathbf{x}, y) dy}{p(\mathbf{x})} = \int y p(y|\mathbf{x}) dy = \mathbb{E}(Y|X)$$

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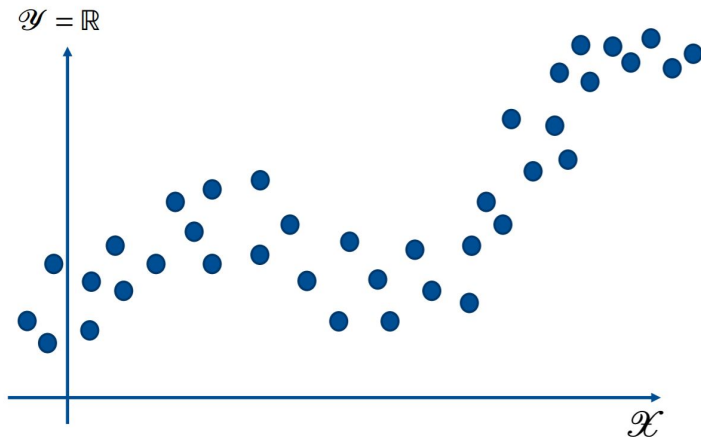
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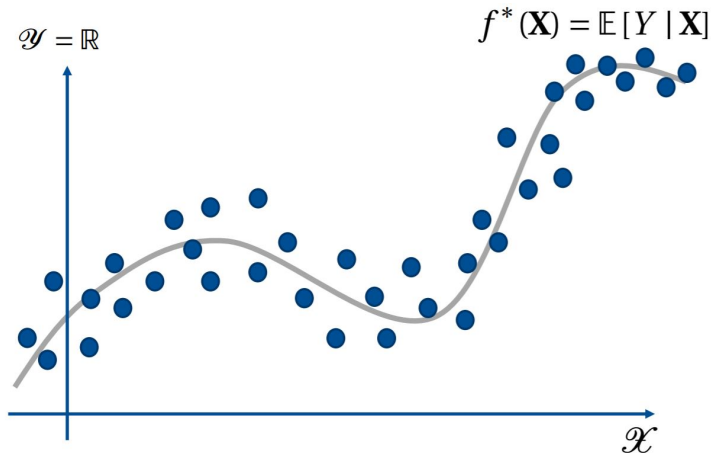
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Alternatives to mean regression

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Example: for a given height, predict the weight such that approximately half of the people with this height would be heavier than this

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- Quantile regression:

Median regression generalized to any other quantile (median is the 50%-quantile)

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- Therefore, the most usual evaluation measure in (mean) regression is **mean squared error (MSE)** on test data:

$$\frac{1}{|Te|} \sum_{(\mathbf{x}, y) \in Te} (\hat{f}(\mathbf{x}) - y)^2$$

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- The advantage of RMSE: easier to interpret because it is measured in the same units as the target variable (whereas MSE is measured in these units squared)

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- The task in mean regression is to predict $f^*(X)$ and the noise cannot be (and should not attempted to be) predicted from the features X

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- Suppose we want to predict a person's weight from height
- What is the noise here?
- Noise is the difference between the actual weight and the average weight among people with the same height

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 - $\mathbf{w}$ - coefficients (coefficient vector)

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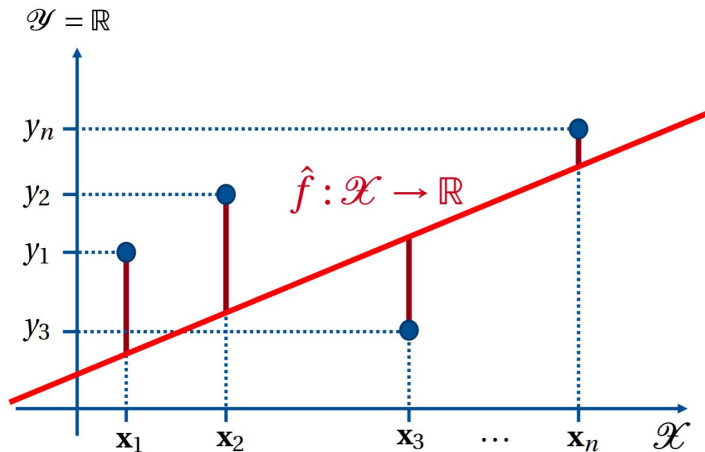
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- One of the popular methods is the Ordinary Least Squares (OLS) method

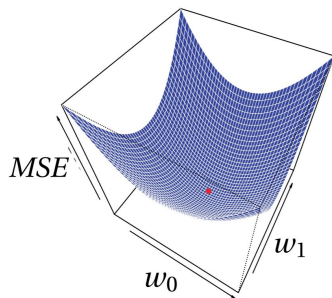
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Ordinary least squares (OLS) regression learns the weights by minimizing MSE on training data:

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Calculating the partial derivatives

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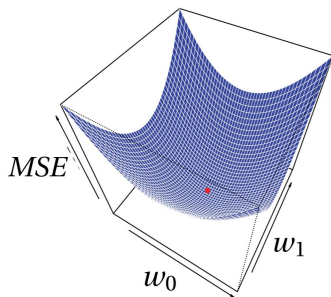
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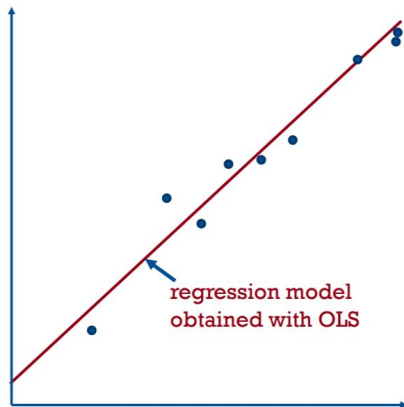
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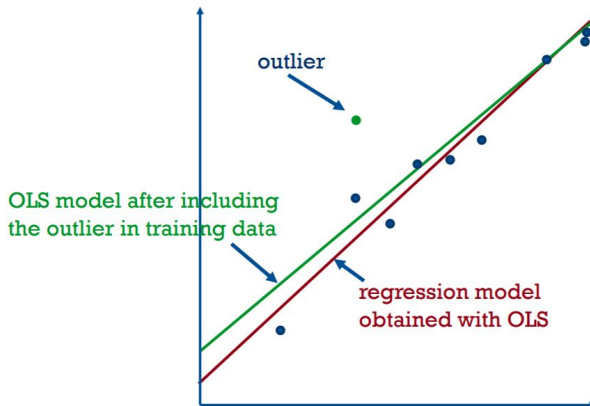
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from now on we will skip the ' for simplicity

OLS in matrix form

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- Denote the training instances by $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n) \in \mathbb{R}^{d+1} \times \mathbb{R}$

OLS in matrix form

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- To solve this we equate the gradient to zero:

$$\frac{\partial (\mathbf{y} - \mathbf{X}\mathbf{w})^T (\mathbf{y} - \mathbf{X}\mathbf{w})}{\partial \mathbf{w}} = 0$$

Calculating the partial derivatives

Based on the properties of matrix derivatives:

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For multivariate OLS there exists a **closed-form solution** (i.e. can be explicitly calculated without numerical optimization):

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Multivariate OLS

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- Con:** otherwise OLS tends to overfit the noise, particularly if there is a lot of noise in the data
- Con:** if many features are collinear (highly correlated) then OLS tends to overfit

What have we learned today?

- ✓ Evaluation and Scoring of Classifiers
- ✓ Main Concepts in Regression
- ✓ Linear Regression
- ✓ Ordinary Least Squares (OLS)